

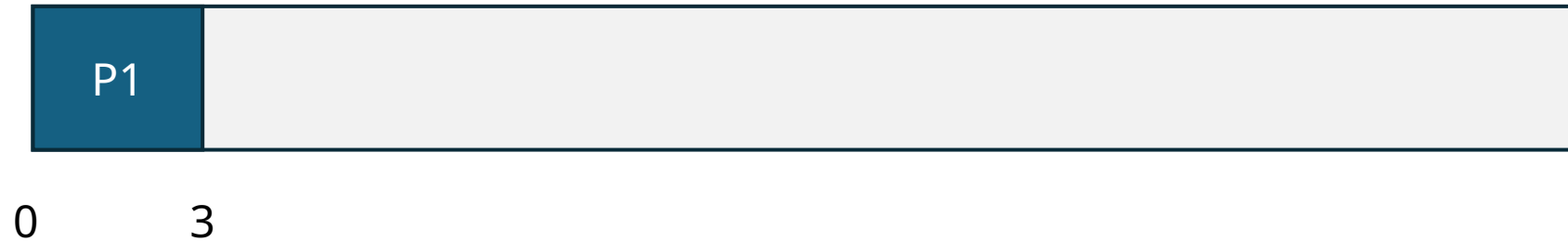
Round Robin Scheduling

Time
quantum =
3

Process	Arrival Time	Burst Time
P1	0	5
P2	1	4
P3	2	2
P4	3	1

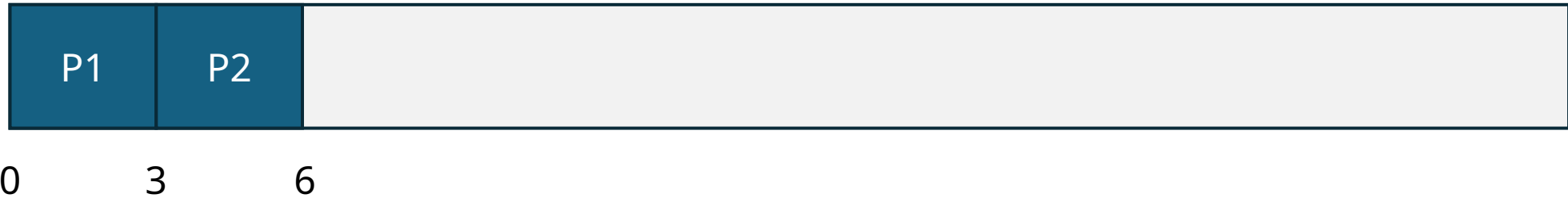
Ready queue: P1

Gantt
chart



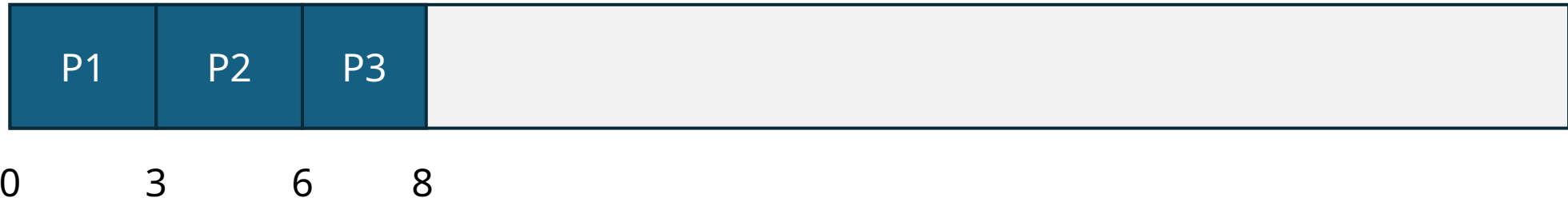
Process	Arrival Time	Burst Time	Remaining Time
P1	0	5	2
P2	1	4	1
P3	2	2	
P4	3	1	

Ready queue: P2 P3 P4 P1



Process	Arrival Time	Burst Time	Remaining Time
P1	0	5	2
P2	1	4	1
P3	2	2	
P4	3	1	

Ready queue: P3 P4 P1 P2



Process	Arrival Time	Burst Time	Remaining Time
P1	0	5	2
P2	1	4	1
P3	2	2	
P4	3	1	

Ready queue: P4 P1 P2



Process	Arrival Time	Burst Time	Remaining Time
P1	0	5	2
P2	1	4	1
P3	2	2	
P4	3	1	

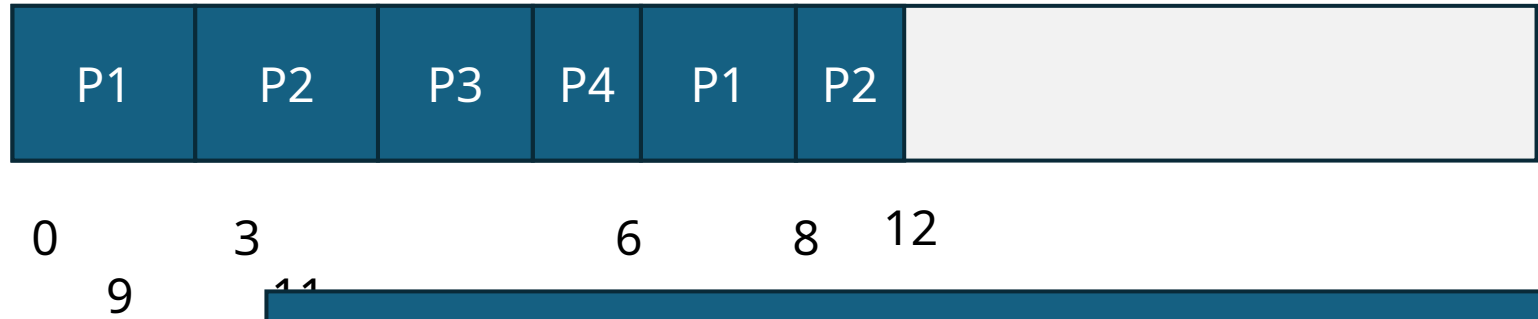
Ready queue: P1 P2



0 3 6 11
 8 9

Process	Arrival Time	Burst Time	Remaining Time
P1	0	5	2
P2	1	4	1
P3	2	2	
P4	3	1	

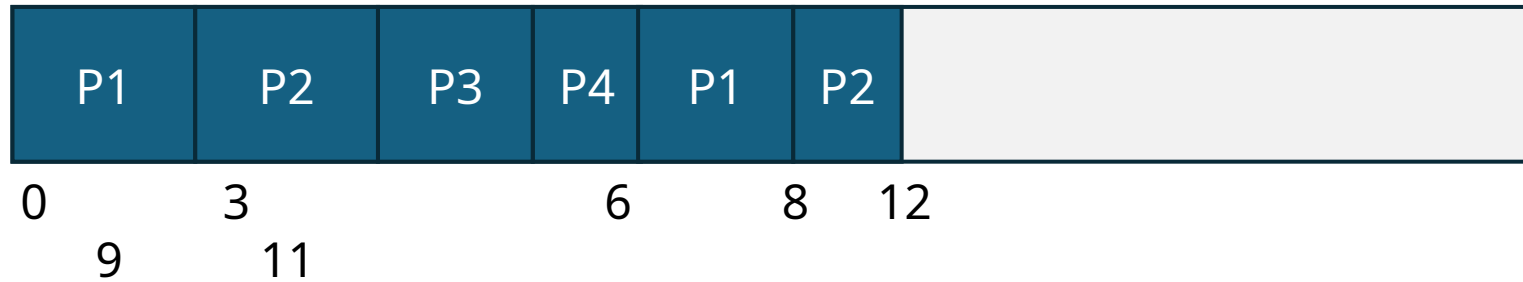
Ready queue: P2



Final Ready Queue

Ready queue:

Calculation of Turnaround Time and Waiting Time



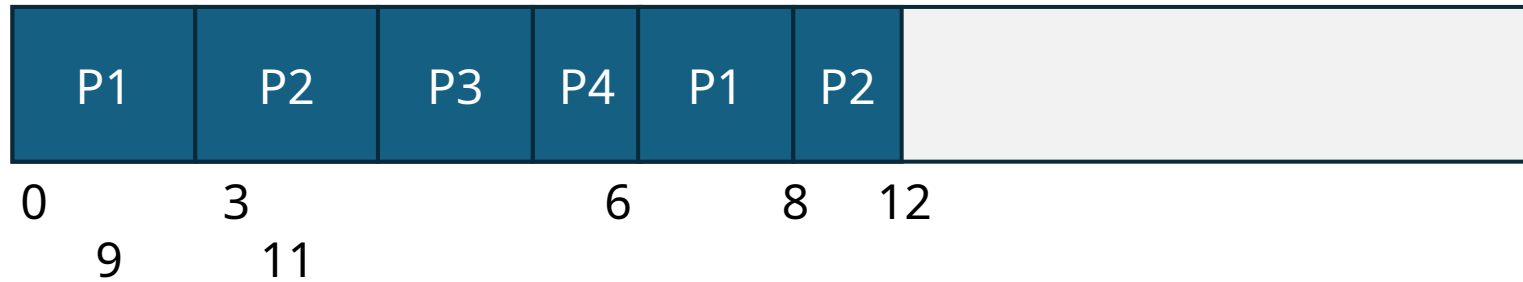
Turnaround time(TA) = Completion time(CT) – Arrival Time(AT)

Waiting Time(WT) = Completion time – Arrival Time – Burst Time
= Turnaround Time – Burst time

Process	AT	CT	TAT
P1	0	11	11-0 = 11
P2	1	12	12-1 = 11
P3	2	8	8-2 = 6
P4	3	9	9-3 = 6

Turnaround Time(TAT)

Calculation of Turnaround Time and Waiting Time



Turnaround time(TA) = Completion time(CT) – Arrival Time(AT)

Waiting Time(WT) = Completion time – Arrival Time – Burst Time
= Turnaround Time – Burst time

Process	BT	TAT	WT
P1	5	11	11-5 = 6
P2	4	11	11-4 = 7
P3	2	6	6-2 = 4
P4	1	6	6-1 = 5

Waiting Time(WT)

Average waiting time(AWT) = (6+7+4+5) / 4 = 5.5

Application of Different CPU Scheduling algorithms

Scheduling Algorithm	Key Idea	Applications
First Come First Served (FCFS)	Processes execute in the order they arrive.	Batch processing systems, print spooling systems, file processing servers.
Shortest Job First (SJF)	Process with the shortest burst time executes first.	Batch systems, scientific computing tasks, systems where job execution time is known.
Shortest Remaining Time First (SRTF)	Preemptive version of SJF; process with the smallest remaining time runs first.	Interactive systems, systems needing minimum turnaround time.
Priority Scheduling	CPU allocated to the highest priority process.	Real-time systems, medical monitoring systems, safety-critical applications.
Round Robin (RR)	Each process gets CPU for a fixed time quantum.	Fair CPU sharing system, time-sharing systems, multi-user operating systems .
Multilevel Queue Scheduling	Processes divided into different queues based on type.	Systems separating system processes, interactive processes, and batch processes.